

# Probability from Observation: Collective Exhaustion and the Observable Extension Theorem

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## Abstract

We prove that a coherent family of observations determines a unique probability measure on the observable  $\sigma$ -algebra. The setting is a directed system of measurable outcome spaces with surjective refinement maps and a compatible family of finitely additive charges on the cylinder algebra. The necessary and sufficient condition for  $\sigma$ -additive extension is *collective exhaustion* (CE): no level can assign persistent mass to events the system as a whole sees as empty. CE is not derivable from any structural condition on the query system — a proved boundary, located precisely by the finite-cofinite counterexample and a model-theoretic argument via Łoś’s theorem.

The result is established by two routes that are not symmetric in their starting assumptions. The *direct Carathéodory route* assumes compatible  $\sigma$ -additive marginals and constructs the global measure directly, with no topological assumptions; CE does not appear as a proof step but identifies, via the CE characterisation theorem, when finitely additive level data can be promoted to this input. The *Stone route* assumes only finite additivity: compactness of the Stone space manufactures  $\sigma$ -additivity on the compact completion, and CE is the support condition that descends the measure back to the instantiation space. Under shared hypotheses the two constructions coincide.

The Stone space  $\text{St}(\mathcal{C})$  is the canonical compact Hausdorff completion of the observable structure: the space of all consistent distinction patterns extending the cylinder algebra.

**2020 Mathematics Subject Classification.** 60A10 (primary); 28A60, 06E15, 28A05 (secondary).

**Keywords.** observable extension theorem, collective exhaustion, Carathéodory extension, Stone duality, Boolean algebra, finitely additive measure,  $\sigma$ -additivity, directed system, projective limit.

## 1 Introduction

The programme of which this paper is a part does not begin with a probability space but with *distinctions*: the acts of observational separation by which an observer organises their experience of a system. State space,  $\sigma$ -algebra, and measure are not primitives — they are what coherent distinctions, pursued to their conclusion, are forced to produce. This paper proves the first step: a coherent family of observations determines a unique probability measure. The further steps — that this measure carries canonical dynamics, that the delay structure of observation (re)constructs the state space, and that finite data can certify (re)construction — are carried out in the companion papers.

What makes the first step possible is a single condition. *Collective exhaustion* (CE): a compatible family of finitely additive charges extends to a  $\sigma$ -additive measure if and only if, whenever a decreasing sequence of events vanishes globally, there exists a level at which the sequence’s mass converges to zero. CE rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit. It is the exact admissibility condition separating observational coherence from genuine probability: a coherent family of charges deserves to be treated as a probability system precisely when it satisfies CE.

The result is established by two routes that are not symmetric in their starting assumptions.

The *Carathéodory route* (§3) is the direct local-to-global extension theorem: given compatible  $\sigma$ -additive probability marginals, it constructs the global observable law directly. Theorem 4.10 identifies CE as the condition under which finitely additive level data can be promoted to the  $\sigma$ -additive marginals this route assumes.

The *Stone route* (§5) is the admissibility route: it assumes only finitely additive compatible charges and derives  $\sigma$ -additivity from compactness. Descending from the Stone space to  $\Omega$  requires two conditions: discriminability ensures the Stone embedding is injective, and CE appears as a support condition — the measure concentrates on the principal ultrafilters, i.e., on the image of  $\Omega$  inside its Stone compactification.

CE is a non-derivable admissibility condition: no first-order structural condition on the query system implies it, as shown by the finite-cofinite counterexample and a model-theoretic argument via Łoś’s theorem (§4). This obstruction is not an artifact of the query-system formalism: in the more primitive first-order language of Boolean algebras with finitely additive charge, countable additivity is likewise not first-order axiomatizable — the need for an extra admissibility condition is logically unavoidable.

The Stone space  $\text{St}(\mathcal{C})$  is the canonical compact Hausdorff completion of  $\Omega$ ; CE’s two faces — algebraic convergence and geometric support — are the same condition seen from inside and outside the algebra (§6).

**Organisation.** Section 2 introduces query systems, cylinder sets, the observable  $\sigma$ -algebra, compatible charges, and the structural hypotheses (Surjective Evaluation, Discriminability). Section 3 gives the direct extension theorem: given compatible  $\sigma$ -additive marginals, the global observable law follows by Carathéodory. Section 4 establishes CE as the admissibility criterion that determines when finitely additive level data warrants that probabilistic status, and proves its non-derivability from structural conditions. Section 5 proves the Stone realization from finitely additive data. Section 6 draws the routes together and discusses the Stone space. The main results have been formalized in Lean 4 using Mathlib; source files are available in the accompanying repository.

## 2 Setup: query systems and charges

### 2.1 Observational structure

Rather than a single  $\sigma$ -algebra of events, we work with a directed system of Boolean algebras representing observable distinctions at increasing resolution; the global  $\sigma$ -algebra arises as their completion.

**Definition 2.1** (Observational structure). An *observational structure* consists of:

- a nonempty partially ordered set  $(\iota, \leq)$ , the *index set*;
- for each  $i \in \iota$ , a measurable space  $(\mathcal{O}_i, \mathcal{O}_i)$ , the *outcome space at level  $i$* ;
- for each  $i \leq j$  in  $\iota$ , a measurable surjection  $\pi_{ij} : \mathcal{O}_j \rightarrow \mathcal{O}_i$ , the *refinement map*,

satisfying  $\pi_{ik} = \pi_{ij} \circ \pi_{jk}$  for all  $i \leq j \leq k$ .

**Definition 2.2** (Instantiation; query system). An *instantiation* of an observational structure  $(\iota, \leq, \{(\mathcal{O}_i, \mathcal{O}_i)\}, \{\pi_{ij}\})$  is a set  $\Omega$  together with maps  $\text{eval}_i : \Omega \rightarrow \mathcal{O}_i$  for each  $i \in \iota$ , the *evaluation maps*, satisfying the *coherence condition*  $\pi_{ij} \circ \text{eval}_j = \text{eval}_i$  for all  $i \leq j$ . A *query system* is an observational structure together with an instantiation.

The index set, outcome spaces, and refinement maps encode the distinctions the observer can make and how finer levels coarsen. Choosing  $\Omega$  is an additional act: it is a set of states realising the abstract structure concretely, and coherence says evaluation is compatible with coarsening.

## 2.2 Canonical instantiation and observables

Every observational structure admits a canonical instantiation: the space of all coherent assignments of outcomes across levels. This identifies  $\Omega$  with the set of all globally consistent observation patterns. Explicitly, the canonical instantiation is the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathcal{O}_i = \left\{ \omega \in \prod_{i \in \iota} \mathcal{O}_i \mid \pi_{ij}(\omega_j) = \omega_i \text{ whenever } i \leq j \right\},$$

with  $\text{eval}_i(\omega) = \omega_i$  and  $\omega \in \Omega$  generic.

**Example 2.3** (Bernoulli observations). Let  $\iota$  be the set of finite subsets of  $\mathbb{N}$ , ordered by inclusion. For  $i \in \iota$ , set  $\mathcal{O}_i = \{0, 1\}^{|i|}$  with its discrete  $\sigma$ -algebra  $\mathcal{O}_i$ , and for  $i \subseteq j$  let  $\pi_{ij} : \mathcal{O}_j \rightarrow \mathcal{O}_i$  be the coordinate projection. An instantiation assigns to each  $\omega$  a consistent family of finite binary observations; the canonical instantiation identifies  $\Omega$  with  $\{0, 1\}^{\mathbb{N}}$ . Cylinder sets correspond to fixing finitely many coordinates, and a compatible family  $\{\nu_i\}$  is precisely the system of finite-dimensional distributions of a Bernoulli process.

**Definition 2.4** (Common coarsenings and common refinements). The index set  $(\iota, \leq)$  admits *common coarsenings* if every pair  $i, j \in \iota$  has a lower bound: there exists  $k \leq i$  and  $k \leq j$ . It admits *common refinements* if every pair has an upper bound: there exists  $k \geq i$  and  $k \geq j$ . It admits *sequential common refinements* if every countable family  $\{i_n\}_{n \geq 1} \subset \iota$  has an upper bound in  $\iota$ .

Sequential common refinements imply common refinements. Common coarsenings ensure cylinder sets form a  $\pi$ -system (Lemma 2.6); sequential common refinements supply the  $\sigma$ -subadditivity step in the Carathéodory route.

**Definition 2.5** (Cylinder sets). For  $i \in \iota$  and  $A \in \mathcal{O}_i$ , the *level- $i$  cylinder with base  $A$*  is

$$\text{Cyl}(i, A) = \{\omega \in \Omega : \text{eval}_i(\omega) \in A\}.$$

The *cylinder family* is  $\mathcal{C} = \{\text{Cyl}(i, A) : i \in \iota, A \in \mathcal{O}_i\}$ , and the *observable  $\sigma$ -algebra* is  $\sigma(\mathcal{C})$ .

**Lemma 2.6** (Cylinder  $\pi$ -system). Assume  $(\iota, \leq)$  admits common coarsenings. Then  $\mathcal{C}$  is a  $\pi$ -system: it is closed under finite intersections.

*Proof.* The coherence condition gives  $\text{Cyl}(i, A) = \text{Cyl}(j, \pi_{ij}^{-1}(A))$  for  $i \leq j$ : the same subset of  $\Omega$  is a cylinder at any finer level. Given  $\text{Cyl}(i, A)$  and  $\text{Cyl}(j, B)$ , let  $k$  be a common lower bound. Then

$$\text{Cyl}(i, A) \cap \text{Cyl}(j, B) = \text{Cyl}(k, \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. □

For each  $i \in \iota$ , let  $B_i$  denote the Boolean algebra of level- $i$  cylinders  $\{\text{Cyl}(i, A) : A \in \mathcal{O}_i\}$ ; then  $\mathcal{C} = \bigcup_{i \in \iota} B_i$ . For  $i \leq j$ , define the *connecting map*

$$\varphi_{ij} : B_i \rightarrow B_j, \quad \varphi_{ij}(\text{Cyl}(i, A)) = \text{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family  $\{B_i, \varphi_{ij}\}$  is a directed system of Boolean algebras.

### 2.3 Charges and hypotheses

**Definition 2.7** (Compatible family). A *charge* on a Boolean algebra  $B$  is a finitely additive function  $\mu : B \rightarrow [0, \infty)$  with  $\mu(\emptyset) = 0$ . A family of charges  $\{\mu_i\}_{i \in \iota}$  with  $\mu_i$  on  $B_i$  is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \quad \text{for all } i \leq j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on  $\mathcal{C}$  by  $\mu(\text{Cyl}(i, A)) := \mu_i(\text{Cyl}(i, A))$ ; compatibility ensures this is well-defined.

The two routes impose different structural conditions on the query system.

**Definition 2.8** (Surjective Evaluation). The query system satisfies *Surjective Evaluation* if  $\text{eval}_i : \Omega \rightarrow \mathcal{O}_i$  is surjective for every  $i \in \iota$ .

Surjective evaluation ensures every outcome at every level is realised by some  $\omega \in \Omega$ ; it implies the refinement maps  $\pi_{ij}$  are surjective and the connecting maps  $\varphi_{ij}$  injective.

**Definition 2.9** (Discriminability). The query system *discriminates points* if for every  $\omega \neq \omega'$  in  $\Omega$  there exists  $E \in \mathcal{C}$  with  $\omega \in E$  and  $\omega' \notin E$ .

Discriminability ensures the Stone embedding is injective; it is required by the Stone route (§5) but not by the Carathéodory route.

## 3 Direct extension from compatible probability marginals

Given compatible  $\sigma$ -additive probability marginals, Carathéodory's theorem yields a global measure directly on the instantiation space. The input is a query system satisfying Surjective Evaluation, sequential common refinements, and common coarsenings, together with a compatible family of  $\sigma$ -additive probability marginals. Section 4 identifies the admissibility condition under which finitely additive level data can be promoted to this input.

### 3.1 Observational determination

**Proposition 3.1** (Observational determination). *Let the query system admit common coarsenings and let  $P, P'$  be probability measures on  $(\Omega, \sigma(\mathcal{C}))$ . If  $(\text{eval}_i)_\# P = (\text{eval}_i)_\# P'$  for all  $i \in \iota$ , then  $P = P'$ .*

*Proof.* The equality of pushforward laws implies  $P$  and  $P'$  agree on every cylinder event. By Lemma 2.6 the cylinder family  $\mathcal{C}$  is a  $\pi$ -system generating  $\sigma(\mathcal{C})$ . The  $\pi$ - $\lambda$  theorem gives  $P = P'$ .  $\square$

### 3.2 Observable content

**Lemma 3.2** (Finite observational content). *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let  $\{\nu_i\}_{i \in \iota}$  be a family with each  $\nu_i \in \mathcal{P}(\mathcal{O}_i)$  compatible under the refinement maps, i.e.  $(\pi_{ij})_\# \nu_j = \nu_i$  for all  $i \leq j$ . Then the assignment*

$$\mu_0(\text{Cyl}(i, A)) := \nu_i(A), \quad A \in \mathcal{O}_i,$$

*defines a well-defined finitely additive content on  $\mathcal{C}$ .*

*Proof. Well-definedness.* Suppose  $\text{Cyl}(i, A) = \text{Cyl}(j, B)$ . Use common refinements to find  $k$  with  $k \geq i$  and  $k \geq j$ . The constraint sets in  $\mathcal{O}_k$  are  $\pi_{ik}^{-1}(A)$  and  $\pi_{jk}^{-1}(B)$ . The cylinder equality and surjective evaluation (surjectivity of  $\text{eval}_k$ ) imply these coincide in  $\mathcal{O}_k$ . Compatibility then gives  $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$ .

*Finite additivity.* For disjoint cylinders at possibly different levels, compress to a common refinement  $k$  and use additivity of  $\nu_k$ .  $\square$

### 3.3 Carathéodory extension theorem

**Theorem 3.3** (Carathéodory observational extension). *Suppose:*

- (i)  $(\iota, \leq)$  admits common coarsenings and sequential common refinements;
- (ii) the query system satisfies *Surjective Evaluation*;
- (iii) each  $\nu_i$  is a probability measure on  $(\mathcal{O}_i, \mathcal{O}_i)$ , and the family is compatible under the refinement maps:  $(\pi_{ij})_{\#} \nu_j = \nu_i$  for all  $i \leq j$ .

Then there exists a unique probability measure  $P$  on  $(\Omega, \sigma(\mathcal{C}))$  such that  $(\text{eval}_i)_{\#} P = \nu_i$  for every  $i \in \iota$ .

*Proof. Step 1: Finite content.* Theorem 3.2 gives a well-defined finitely additive content  $\mu_0$  on  $\mathcal{C}$ .

*Step 2:  $\sigma$ -subadditivity.* Let  $\text{Cyl}(i, B) \subseteq \bigcup_{n \geq 1} \text{Cyl}(i_n, A_n)$ . Apply sequential common refinements to  $\{i, i_1, i_2, \dots\}$  to find a single index  $k$  with  $k \geq i$  and  $k \geq i_n$  for all  $n$ . Compress each cylinder to level  $k$ :

$$\text{Cyl}(i, B) = \text{Cyl}(k, E), \quad \text{Cyl}(i_n, A_n) = \text{Cyl}(k, E_n),$$

where  $E = \pi_{ik}^{-1}(B)$  and  $E_n = \pi_{i_n k}^{-1}(A_n)$ . Surjective evaluation and the cylinder inclusion in  $\Omega$  imply  $E \subseteq \bigcup_n E_n$  in  $\mathcal{O}_k$ . Therefore

$$\mu_0(\text{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\text{Cyl}(i_n, A_n)),$$

using  $\sigma$ -subadditivity of the single measure  $\nu_k$ .

*Step 3: Carathéodory extension.* The family  $\mathcal{C}$  is a set semiring generating  $\sigma(\mathcal{C})$ . It is closed under finite intersections by Lemma 2.6. Moreover, if  $E = \text{Cyl}(i, A)$  and  $F = \text{Cyl}(j, B)$ , choose  $k$  with  $k \geq i, j$ ; then

$$E = \text{Cyl}(k, \pi_{ik}^{-1}(A)), \quad F = \text{Cyl}(k, \pi_{jk}^{-1}(B)),$$

so

$$E \setminus F = \text{Cyl}(k, \pi_{ik}^{-1}(A) \setminus \pi_{jk}^{-1}(B)) \in \mathcal{C}.$$

Thus  $\mathcal{C}$  is closed under relative differences, confirming the semiring property. Since  $\mu_0$  is a finitely additive content on  $\mathcal{C}$  that is  $\sigma$ -subadditive for countable covers by cylinder sets (Step 2), the semiring form of Carathéodory's extension theorem yields a measure  $P$  on  $(\Omega, \sigma(\mathcal{C}))$  extending  $\mu_0$  (Bogachev [1], Vol. I, §1.5, Thm. 1.5.?).

*Step 4: Marginal recovery and normalization.* For any  $A \in \mathcal{O}_i$ ,  $P(\text{Cyl}(i, A)) = \mu_0(\text{Cyl}(i, A)) = \nu_i(A)$ , so  $(\text{eval}_i)_{\#} P = \nu_i$ . Setting  $A = \mathcal{O}_i$  gives  $P(\Omega) = 1$ .

*Uniqueness.* Two probability measures with the same evaluation marginals agree on  $\mathcal{C}$ , hence on  $\sigma(\mathcal{C})$  by Theorem 3.1.  $\square$

**Remark 3.4** (Relation to Kolmogorov's extension theorem). Kolmogorov's theorem [7] constructs a probability measure on  $S^T$  from consistent finite-dimensional distributions. Theorem 3.3 plays the same role for query systems: the instantiation  $\Omega = \varprojlim \mathcal{O}_i$  replaces the product  $S^T$ , compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure  $P$  is built directly on the measurable structure of  $\Omega$  by Carathéodory, without any appeal to tightness or compactness.

## 4 Collective Exhaustion

Section 3 showed that compatible  $\sigma$ -additive marginals extend directly to a global observable law. This section identifies the admissibility condition that determines when finitely additive level data deserves that probabilistic status.

**Definition 4.1** (Collective Exhaustion). A charge  $\mu$  on  $\mathcal{C}$  (the common extension of a compatible family  $\{\mu_i\}$ ) satisfies *collective exhaustion* (CE) if: whenever  $E_1 \supseteq E_2 \supseteq \dots$  in  $\mathcal{C}$  and  $\bigcap_n E_n = \emptyset$ , we have  $\mu(E_n) \rightarrow 0$ .

CE is a valuation-layer condition: it constrains how charges behave, not the Boolean-algebraic index structure. The following example shows it is not automatic.

**Example 4.2** (Finite-cofinite obstruction). Let  $\mathbb{N}$  carry the finite-cofinite algebra  $\mathcal{E}$  where  $A \in \mathcal{E}$  if  $A$  is finite or  $A^c$  is finite. Define a normalized charge by  $\mu_0(A) = 0$  if  $A$  is finite and  $\mu_0(A) = 1$  if  $A$  is cofinite.

Disjoint sets in  $\mathcal{E}$  are either both finite or one finite and one cofinite (two disjoint cofinite sets cannot exist), so finite additivity holds.

The sequence  $E_n = \{n, n+1, \dots\}$  satisfies  $E_n \downarrow \emptyset$ , yet each  $E_n$  is cofinite, so  $\mu_0(E_n) = 1$  for all  $n$ : mass is not exhausted. No  $\sigma$ -additive extension exists.

We work first at the level of an abstract directed system of Boolean algebras, writing  $\mathcal{E}_i$  for the algebra at level  $i$  and  $\ell_i$  for its charge; the query-system interpretation returns in §5.

### 4.1 Single-algebra extension

**Definition 4.3** (Boolean charge space). A *Boolean charge space* is a pair  $(\mathcal{E}, \ell)$  where  $\mathcal{E}$  is a Boolean algebra of subsets of a set  $O$  and  $\ell : \mathcal{E} \rightarrow [0, 1]$  is a normalized finitely-additive valuation. No  $\sigma$ -algebra and no topology are assumed on  $O$ .

By Stone's representation theorem [11],  $\mathcal{E}$  is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space  $S(\mathcal{E})$ , and  $\sigma(\mathcal{E})$  is the Baire  $\sigma$ -algebra of  $S(\mathcal{E})$ . The sets in  $\sigma(\mathcal{E}) \setminus \mathcal{E}$  are Baire events that countable operations can reach but no single observable distinction can name.

**Lemma 4.4** (Extension criterion). *A normalized finitely-additive  $\ell : \mathcal{E} \rightarrow [0, 1]$  extends to a  $\sigma$ -additive measure on  $\sigma(\mathcal{E})$  if and only if  $\ell$  is continuous at  $\emptyset$ : for every decreasing sequence  $E_1 \supseteq E_2 \supseteq \dots$  in  $\mathcal{E}$  with  $\bigcap_n E_n = \emptyset$  in  $\sigma(\mathcal{E})$ , we have  $\ell(E_n) \rightarrow 0$ . The extension is then unique.*

*Proof.* Standard; see Halmos [5] or Fremlin [4] Vol. 1. Continuity at  $\emptyset$  is used in the Carathéodory extension argument to pass from finite additivity to  $\sigma$ -additivity on  $\sigma(\mathcal{E})$ .  $\square$

The condition in Lemma 4.4 is not verifiable within  $\mathcal{E}$  alone: confirming  $\bigcap_n E_n = \emptyset$  requires access to  $\sigma(\mathcal{E})$ . In a directed system, one might hope a finer level  $j \geq i$  could witness the emptiness level  $i$  cannot see — but the same obstruction reappears: finite additivity at  $j$  does not force continuity at  $\emptyset$  for sequences in  $\sigma(\mathcal{E}_j)$ . Section 4.2 shows the directed-system structure alone does not provide it.

### 4.2 Nontrivial refinement

**Definition 4.5** (Nontrivial refinement). Let  $\pi : O_j \rightarrow O_i$  be a surjective refinement map, inducing the Boolean algebra homomorphism  $\pi^{-1} : \mathcal{E}_i \rightarrow \mathcal{E}_j$ . The refinement is *nontrivial* if there exists  $A \in \mathcal{E}_j$  with  $A \notin \pi^{-1}(\mathcal{E}_i)$ .

**Proposition 4.6** (Nontrivial refinement introduces new events). *If  $j \geq i$  is a nontrivial refinement, then  $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$ . Equivalently, a level  $i$  at which  $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$  for every  $j \geq i$  is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

*Proof.* Nontriviality asserts the existence of  $A \in \mathcal{E}_j$  with  $A \notin \pi^{-1}(\mathcal{E}_i)$ , directly giving the strict inclusion. The contrapositive is the stated equivalence.  $\square$

**Lemma 4.7** (Strict enlargement under  $\sigma$ -closure). *Let  $\mathcal{E}$  be a Boolean algebra of subsets of a countably infinite set  $O$  that is not a  $\sigma$ -algebra. Then  $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$ .*

*Proof.* If  $\mathcal{E}$  is not a  $\sigma$ -algebra, there exists a countable family  $\{A_n\} \subset \mathcal{E}$  with  $\bigcup_n A_n \notin \mathcal{E}$ . By definition  $\bigcup_n A_n \in \sigma(\mathcal{E})$ , so it lies in  $\sigma(\mathcal{E}) \setminus \mathcal{E}$ .  $\square$

### 4.3 CE characterises $\sigma$ -additivity

We now state the directed-system version of CE and prove the main equivalence.

**Definition 4.8** (Collectively exhaustive family). A compatible family  $\{\ell_i\}$  of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every  $i \in \mathcal{I}$  and every decreasing sequence  $E_1 \supseteq E_2 \supseteq \dots$  in  $\mathcal{E}_i$  with  $\bigcap_n E_n = \emptyset$ , there exists  $j \geq i$  such that  $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$ .

Here  $\pi_{ij} : O_j \rightarrow O_i$  is the refinement map ( $j$  finer than  $i$ ), so  $\pi_{ij}^{-1}(E_n) \subseteq O_j$  is the preimage of  $E_n$  at the finer level. No level may assign persistent mass to events the system as a whole sees as empty.

The following proposition constructs the minimal directed system in which the index-layer hypotheses hold while the finite-cofinite charge still fails  $\sigma$ -additive extension. The construction is a deliberate stress test, not a naturally arising observational system.

**Proposition 4.9** (Index-layer conditions do not force  $\sigma$ -additivity). *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails  $\sigma$ -additive extension at every level.*

*Proof.* We take the simplest sequentially upper-directed index set: adjoin a single terminal element  $\omega$  to  $\mathbb{N}$ , setting  $n \leq \omega$  for all  $n \in \mathbb{N}$ . Take all outcome spaces to be  $\mathbb{N}$ , all event algebras to be the finite-cofinite algebra  $\mathcal{E}$ , all refinement maps to be the identity, and each  $\ell_i$  to be the finite-cofinite charge  $\ell$  of Example 4.2. Compatibility holds trivially. The system satisfies every structural condition. By Example 4.2,  $\ell$  is not  $\sigma$ -additive and admits no  $\sigma$ -additive extension; hence every level fails  $\sigma$ -additive extension.  $\square$

**Theorem 4.10** (CE Characterisation). *Let  $\{\ell_i\}$  be a normalized compatible family of finitely-additive charges on a directed system indexed by  $\mathcal{I}$ . The following are equivalent:*

- (i) *The family is collectively exhaustive.*
- (ii)  *$\ell_i$  extends uniquely to a  $\sigma$ -additive measure on  $\sigma(\mathcal{E}_i)$  for every  $i \in \mathcal{I}$ .*

*Proof.* (i)  $\Rightarrow$  (ii). Fix  $i \in \mathcal{I}$  and a decreasing sequence  $E_n \searrow \emptyset$  in  $\mathcal{E}_i$ . By CE, find  $j \geq i$  with  $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$ . By compatibility,  $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$ . So  $\ell_i$  is continuous at  $\emptyset$ . Lemma 4.4 gives the unique  $\sigma$ -additive extension.

(ii)  $\Rightarrow$  (i). Suppose  $\ell_i$  extends to  $\mu_i$  for every  $i$ . Let  $E_n \searrow \emptyset$  in  $\mathcal{E}_i$ . Then  $\ell_i(E_n) = \mu_i(E_n) \rightarrow \mu_i(\emptyset) = 0$  by  $\sigma$ -additivity of  $\mu_i$  and normalization. Take  $j = i$ .  $\square$

The question “do the structural conditions on a directed system force  $\sigma$ -additivity?” is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 4.9 separates them; Theorem 4.10 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§3).

#### 4.4 CE is non-derivable

The Yosida–Hewitt decomposition [12] identifies the obstruction Proposition 4.9 has exposed. Every finitely additive probability decomposes uniquely as  $\ell = \ell_c + \ell_p$ , where  $\ell_c$  is  $\sigma$ -additive and  $\ell_p$  is *purely finitely additive*. The finite-cofinite content has  $\ell_p = \ell$ . By Theorem 4.10, CE holds precisely when  $\ell_p = 0$ . The obstruction is therefore not a feature of any particular structural hypothesis: it is the persistence of a purely finitely additive component that no first-order structural condition can eliminate. Proposition 4.11 makes this precise.

**Proposition 4.11** (CE non-derivability). *Collective exhaustion is not implied by any structural condition on a query system.*

- (i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*
- (ii) (Universal case.) *No condition  $\Phi$  expressible in the first-order language of query systems implies CE.*

*Proof.* The particular case is Proposition 4.9: the finite-cofinite content satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

*Step 1.* A first-order condition on query systems is preserved under ultraproducts by Łoś’s theorem [8]: if  $\Phi$  holds in every member of a family of query systems, it holds in their ultraproduct.

*Step 2.* Since the finite-cofinite construction works on any countably infinite set, we may realise it on  $\mathbb{Q}$ . Fix an enumeration  $q : \mathbb{N} \rightarrow \mathbb{Q}$ , let  $\mathcal{U}$  be a nonprincipal ultrafilter on  $\mathbb{N}$  extending the cofinite filter, and let  $\delta_n$  be the point mass at  $q(n) \in \mathbb{Q}$ . Each  $\delta_n$  is a normalized  $\sigma$ -additive content satisfying every structural condition including CE. The ultraproduct  $\prod_{\mathcal{U}} \delta_n$  assigns to  $E \subseteq \mathbb{Q}$  the value  $\lim_{\mathcal{U}} \mathbf{1}_{q(n) \in E}$ .

If  $E$  is finite then  $q(n) \in E$  holds for only finitely many  $n$ , so the ultralimit is 0. If  $E$  is cofinite then  $q(n) \in E$  holds for cofinitely many  $n$ , so the ultralimit is 1, since  $\mathcal{U}$  extends the cofinite filter. The ultraproduct is therefore exactly the finite-cofinite content  $\ell$  of Example 4.2 on  $\mathbb{Q}$ .

*Step 3.* By Step 1, any universally valid first-order condition  $\Phi$  holds in the ultraproduct whenever it holds in each  $\delta_n$ . But by Step 2 the ultraproduct is exactly  $\ell$ , and  $\ell$  fails CE. Therefore no first-order structural condition can imply CE.  $\square$

The ultrafilter-based content is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty. CE is the admissibility condition under which observational coherence furnishes probability, not a structural consequence of the query system.

**Remark 4.12** (Metatheoretic scope). The obstruction is not an artifact of the query-system language. In the more primitive setting of Boolean algebras equipped with a normalized finitely additive charge, the same conclusion holds: no first-order theory in that language characterizes those algebras whose charge extends to a  $\sigma$ -additive measure. Equivalently, countable additivity is not first-order axiomatizable among finitely additive probability Boolean algebras [9].

The proof is the same ultraproduct construction, now stated in full generality. Each Dirac charge  $\delta_n$  on the finite-cofinite algebra of  $\mathbb{Q}$  is  $\sigma$ -additive, so any putative first-order theory characterizing  $\sigma$ -additive extensibility would be satisfied by every  $\delta_n$ . Their ultraproduct (along a nonprincipal ultrafilter extending the cofinite filter) yields the finite-cofinite charge  $\ell$ , which is purely finitely additive and fails  $\sigma$ -additivity. By Łoś’s theorem, any first-order theory satisfied by all  $\delta_n$  is satisfied by the ultraproduct — but the ultraproduct fails  $\sigma$ -additivity. Contradiction.

The philosophical upshot is not that one foundational position wins. It is that finite coherence underdetermines probability in a logically precise sense: no finitary first-order description of observational structure can close the gap. The need for some extra admissibility principle is unavoidable. CE is the condition adopted here; other foundational approaches may supply different extra structure, but some such structure is required. Foundational disputes about probability are not terminological; they are disputes about which extra structure licenses the passage from finite coherence to probability.

## 5 Stone realization from finitely additive data

The Carathéodory route may suggest that  $\sigma$ -additivity must be assumed from the outset. The second route shows it can instead be derived. Starting only from finitely additive charges, compactness of the Stone space produces a  $\sigma$ -additive measure on the compact completion; CE reappears as the support condition that returns this measure to the realised instantiation space.

The proof proceeds in four steps. §5.1 identifies the Stone space of the direct limit with the inverse limit of individual Stone spaces and applies Choksi's theorem. §5.3 identifies the pullback of the Baire algebra of the Stone space with the observable  $\sigma$ -algebra. §5.4 shows CE forces the measure to concentrate on the image of  $\Omega$ .

### 5.1 Stone duality for the direct limit

**Lemma 5.1** (Injectivity of connecting maps). *If the query system satisfies Surjective Evaluation, then  $\varphi_{ij} : B_i \rightarrow B_j$  is an injective Boolean algebra homomorphism for every  $i \leq j$ .*

*Proof.* Surjective evaluation and the coherence condition give  $\pi_{ij}(\text{eval}_j(\omega)) = \text{eval}_i(\omega)$  for all  $\omega$ , so  $\text{im}(\pi_{ij}) \supseteq \text{im}(\text{eval}_i) = \mathcal{O}_i$ : the refinement maps are surjective. A surjective map  $\pi_{ij}$  has injective preimage map  $\pi_{ij}^{-1}$  on subsets, so  $\varphi_{ij}(\text{Cyl}(i, A)) = \text{Cyl}(j, \pi_{ij}^{-1}(A))$  is injective on generators, hence on  $B_i$ .  $\square$

**Proposition 5.2** (Stone duality for the direct limit). *The Stone space of the direct limit satisfies  $\text{St}(\bigcup_i B_i) \cong \varprojlim_i \text{St}(B_i)$ .*

*Proof.* By Lemma 5.1, the connecting maps  $\varphi_{ij}$  are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor  $\text{St}$  converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [6, II.4]. Applied to  $\bigcup_i B_i$  with its injective system, this gives  $\text{St}(\bigcup_i B_i) \cong \varprojlim_i \text{St}(B_i)$ .  $\square$

### 5.2 The inverse limit measure

**Proposition 5.3** (Inverse limit measure). *Let  $\{\mu_i\}$  be a compatible family of normalised charges on  $\{B_i\}$ . Then there exists a unique probability measure  $\hat{\mu}$  on  $\mathcal{B}(\varprojlim_i \text{St}(B_i))$  whose pushforward along each projection  $\varprojlim_i \text{St}(B_i) \rightarrow \text{St}(B_j)$  agrees with the Borel measure on  $\text{St}(B_j)$  corresponding to  $\mu_j$ .*

*Proof. Step 1: Single-algebra correspondence.* For each  $i$ , Stone duality identifies  $B_i$  with the clopen algebra  $\text{Clop}(\text{St}(B_i))$ . A charge  $\mu_i$  on  $B_i$  thus defines a finitely additive set function on the clopens of the compact Hausdorff space  $\text{St}(B_i)$ . Since  $\text{St}(B_i)$  is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [2], §6, or [5], §§53–54. Call this measure  $\hat{\mu}_i$  on  $\text{St}(B_i)$ .

*Step 2: Compatibility.* The compatibility condition  $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$  translates, under Stone correspondence, to  $\hat{\mu}_i$  being the pushforward of  $\hat{\mu}_j$  along the bonding map  $\text{St}(B_j) \rightarrow \text{St}(B_i)$ . The family  $\{\hat{\mu}_i\}$  is compatible on the inverse system  $\{\text{St}(B_i)\}$ .

*Step 3: Extension to the inverse limit.* Each  $\text{St}(B_i)$  is compact Hausdorff. The inverse limit  $\varprojlim_i \text{St}(B_i)$  is compact Hausdorff. By [3, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives  $\hat{\mu}$  on  $\mathcal{B}(\varprojlim_i \text{St}(B_i))$ .  $\square$

### 5.3 Structural identification

Let  $\text{pure} : \Omega \rightarrow \text{St}(\mathcal{C})$  denote the Stone embedding, which sends  $\omega \in \Omega$  to its principal ultrafilter  $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ . For  $E \in \mathcal{C}$ , write  $[E] = \{u \in \text{St}(\mathcal{C}) : E \in u\}$  for the corresponding basic clopen in  $\text{St}(\mathcal{C})$ .

**Proposition 5.4** (Structural identification). *Suppose the query system discriminates points. Then*

$$\text{pure}^{-1}(\mathcal{B}_{\text{Ba}}(\text{St}(\mathcal{C}))) = \sigma(\mathcal{C}),$$

where  $\mathcal{B}_{\text{Ba}}$  denotes the Baire  $\sigma$ -algebra.

*Proof.* ( $\supseteq$ ): For any  $E = \text{Cyl}(i, A) \in \mathcal{C}$ ,  $\text{pure}^{-1}([E]) = \{\omega : E \in \text{pure}(\omega)\} = \{\omega : \omega \in E\} = E$ . Since  $[E]$  is clopen, hence Baire, every cylinder set is in  $\text{pure}^{-1}(\mathcal{B}_{\text{Ba}})$ , and therefore  $\sigma(\mathcal{C}) \subseteq \text{pure}^{-1}(\mathcal{B}_{\text{Ba}})$ .

( $\subseteq$ ): The Stone space  $\text{St}(\mathcal{C})$  is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets  $[E]$  for  $E \in \mathcal{C}$  [6, I.3.2], so the Baire  $\sigma$ -algebra of  $\text{St}(\mathcal{C})$  is  $\sigma(\{[E] : E \in \mathcal{C}\})$ . For any Baire set  $V$ ,  $\text{pure}^{-1}(V) \in \sigma(\{\text{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$ .

**Injectivity.** Discriminability (Definition 2.9) implies  $\text{pure}$  is injective: if  $\omega \neq \omega'$ , there exists  $E \in \mathcal{C}$  with  $\omega \in E$  and  $\omega' \notin E$ , so  $E \in \text{pure}(\omega)$  but  $E \notin \text{pure}(\omega')$ . Injectivity ensures the pullback does not conflate distinct events.  $\square$

**Remark 5.5.** On a compact Hausdorff space the Baire  $\sigma$ -algebra is contained in the Borel  $\sigma$ -algebra, with equality when the space is second-countable. We work throughout with the Baire  $\sigma$ -algebra on  $\text{St}(\mathcal{C})$ ; this suffices because the measure  $\hat{\mu}$  constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

### 5.4 CE as support condition

**Proposition 5.6** (Support condition). *Let  $\mu$  be a charge on  $\mathcal{C}$  satisfying CE. Then the corresponding Baire measure  $\hat{\mu}$  on  $\text{St}(\mathcal{C})$  is supported on  $\text{pure}(\Omega)$ , the set of principal ultrafilters.*

*Proof.* By the Yosida–Hewitt theorem [12, 10], every bounded charge on a Boolean algebra decomposes uniquely as  $\mu = \mu_c + \mu_p$ , where  $\mu_c$  is countably additive and  $\mu_p$  is purely finitely additive. Under Stone duality,  $\mu_c$  corresponds to mass on the principal ultrafilters  $\text{pure}(\Omega) \subset \text{St}(\mathcal{C})$ , while  $\mu_p$  corresponds to mass on the non-principal ultrafilters [10], Ch. 10.

CE is equivalent to  $\mu_p = 0$ : CE requires  $\mu(E_n) \rightarrow 0$  whenever  $E_n \searrow \emptyset$ , which is exactly the condition that the purely finitely additive part vanishes (Theorem 4.10). Therefore  $\hat{\mu}_p = 0$  and  $\hat{\mu}$  is supported on  $\text{pure}(\Omega)$ .  $\square$

### 5.5 The Stone observational extension

**Theorem 5.7** (Stone observational extension). *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let  $\{\mu_i\}$  be a compatible family of normalised charges on  $\{B_i\}$  satisfying CE. Then there exists a unique probability measure  $P$  on  $(\Omega, \sigma(\mathcal{C}))$  such that*

$$P(\text{Cyl}(i, A)) = \mu_i(\text{Cyl}(i, A)) \quad \text{for all } i \in \iota \text{ and } A \in \mathcal{O}_i.$$

*Proof.* The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\text{\S 5.2, Step 1}} \{\hat{\mu}_i\} \xrightarrow{\text{\S 5.2, Step 3}} \hat{\mu} \text{ on } \varprojlim_i \text{St}(B_i) \xrightarrow{\text{B1}} P \text{ on } \sigma(\mathcal{C}).$$

*Existence.* By Proposition 5.2,  $\text{St}(\mathcal{C}) \cong \varprojlim_i \text{St}(B_i)$ . By Proposition 5.3, the compatible family  $\{\mu_i\}$  extends to a Baire probability measure  $\hat{\mu}$  on  $\varprojlim_i \text{St}(B_i) \cong \text{St}(\mathcal{C})$ .

By Proposition 5.6,  $\hat{\mu}$  is supported on  $\text{pure}(\Omega)$ . Since  $\text{pure}$  is injective (discriminability), the pushforward  $P = (\text{pure})_* \hat{\mu}$  — the image measure under  $\text{pure}^{-1}$  on the support — is a well-defined probability measure on  $\Omega$ .

By Proposition 5.4,  $\text{pure}^{-1}$  maps Baire sets in  $\text{St}(\mathcal{C})$  to sets in  $\sigma(\mathcal{C})$ , so  $P$  is defined on  $\sigma(\mathcal{C})$ . For any cylinder set  $E = \text{Cyl}(i, A)$ ,

$$P(E) = \hat{\mu}([E]) = \mu_i(\text{Cyl}(i, A)).$$

*Uniqueness.* Two probability measures on  $(\Omega, \sigma(\mathcal{C}))$  agreeing on all cylinder sets are equal by Theorem 3.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to surjective evaluation, discriminability, and CE.  $\square$

It may appear that compactness of the Stone space eliminates the need for CE. Compactness ensures  $\hat{\mu}$  on  $\text{St}(\mathcal{C})$  is  $\sigma$ -additive, but  $\hat{\mu}$  lives on the Stone space, not on  $\Omega$ . Descent to  $\Omega$  requires that no mass escapes to non-principal ultrafilters, and that is precisely CE (Proposition 5.6). In the Stone picture, CE is not an additional hypothesis but a *support condition*: the measure concentrates on the image of  $\Omega$  inside its Stone compactification.

## 6 The bridge

### 6.1 Two routes, one measure

Under shared hypotheses — surjective evaluation, discriminability, and  $\sigma$ -additive compatible marginals — both routes produce the same measure.

**Proposition 6.1** (Coincidence of routes). *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let  $\{\nu_i\}$  be a compatible family of probability measures. Let  $P^C$  be the measure produced by Theorem 3.3 and  $P^S$  the measure produced by Theorem 5.7. Then  $P^C = P^S$ .*

*Proof.* Both measures agree on every cylinder set:  $P^C(\text{Cyl}(i, A)) = \nu_i(A) = P^S(\text{Cyl}(i, A))$ . By observational determination (Theorem 3.1), they are equal on  $\sigma(\mathcal{C})$ .  $\square$

The two routes answer different questions on the way to the same measure. The Carathéodory route answers the extension question: given compatible probability marginals, build the global observable law directly. The Stone route answers the admissibility question: starting from finitely additive charges, identify CE as the condition licensing  $\sigma$ -additive extension, then realise it via compactness. That both produce the same measure is not obvious from the constructions; it is a consequence of uniqueness (Proposition 3.1).

### 6.2 CE's two faces

CE appears in the two routes in structurally different forms, yet it is the same condition.

In the Carathéodory route, CE does not appear as a proof step: the route begins from already  $\sigma$ -additive marginals. Theorem 4.10 identifies CE as the algebraic condition explaining when finitely additive level charges deserve to be treated as probability measures — when they do not assign persistent mass to events the refinement hierarchy sees as empty. CE is thus the admissibility criterion whose satisfaction licenses the Carathéodory route's input.

In the Stone route, CE appears as a *topological* condition: the Baire measure  $\hat{\mu}$  on the Stone space must be supported on  $\text{pure}(\Omega)$ , the image of the sample space inside its compact compactification. Mass on non-principal ultrafilters corresponds exactly to the purely finitely additive part  $\mu_p$ , and  $\text{CE} \Leftrightarrow \mu_p = 0$  (Proposition 5.6). Here, CE is a *geometric* condition: the observer’s measure does not escape to the “phantom” points of the Stone compactification.

The Yosida–Hewitt decomposition bridges the two faces: it identifies the purely finitely additive part of  $\mu$  with the mass escaped to non-principal ultrafilters. CE says both things at once: algebraic convergence is honest, and the geometric measure is confined to  $\Omega$ .

The two routes are therefore not symmetric alternatives. In the Carathéodory route,  $\sigma$ -additivity is assumed from the outset; CE is the admissibility criterion that explains when finitely additive data warrants that assumption. In the Stone route, compactness manufactures  $\sigma$ -additivity on the compact completion, and CE is the support condition that descends it to  $\Omega$ . In both cases CE does the same work: no mass escapes the realised observational world.

The metatheoretic scope of this result matters for how the paper’s conclusion should be read. The need for CE is not explained by any feature of the query-system formalism, nor by any particular choice of index structure. As Remark 4.12 records, it reflects a genuine expressive limitation of finitary first-order description: finite coherence can be encoded, but the countable-exhaustion behaviour required for probability cannot be derived from it. The two routes show two ways of bridging this unavoidable gap: the Carathéodory route begins from local laws that already carry probabilistic status, while the Stone route bridges it through compactness and the support condition. What CE names, in both cases, is the same gap.

### 6.3 The Stone space as compact completion

The Stone space  $\text{St}(\mathcal{C})$ , constructed as a technical device for measure extension, is the canonical compact Hausdorff completion of  $\Omega$  for the observable topology. The embedding  $\text{pure} : \Omega \rightarrow \text{St}(\mathcal{C})$  is dense — every nonempty basic clopen  $[E]$  meets  $\text{pure}(\Omega)$  — and the Stone space is uniquely determined by  $\mathcal{C}$  up to homeomorphism: the compactification already contained in the Boolean algebra of cylinder sets.

Every point in  $\text{St}(\mathcal{C})$  is an ultrafilter: a maximal consistent assignment of yes/no answers to every observable distinction. Principal ultrafilters correspond to actual sample-space points; non-principal ultrafilters are ideal limit points — consistent distinction patterns the refinement system coheres with, but that no element of  $\Omega$  realises. The Stone space is the *space of all consistent distinction patterns*: the completion of the observer’s world to a compact whole.

CE closes the loop. It requires the measure  $P$ , when extended to  $\text{St}(\mathcal{C})$  via  $\text{pure}$ , to remain concentrated on the realised part: no mass escapes to the phantom ultrafilters. Probability remains confined to the realised world however far the algebraic completion extends. The Stone topology, generated by all cylinder clopens, is the coarsest topology making every observable distinction continuous.

For a dynamical system  $(X, T, \mu)$  with observation function  $h$ , the time-delay cylinder algebras form a directed system of exactly the kind treated here. Whether the Stone space of the resulting observable algebra recovers the state space  $X$  depends on conditions developed in the companion papers.

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